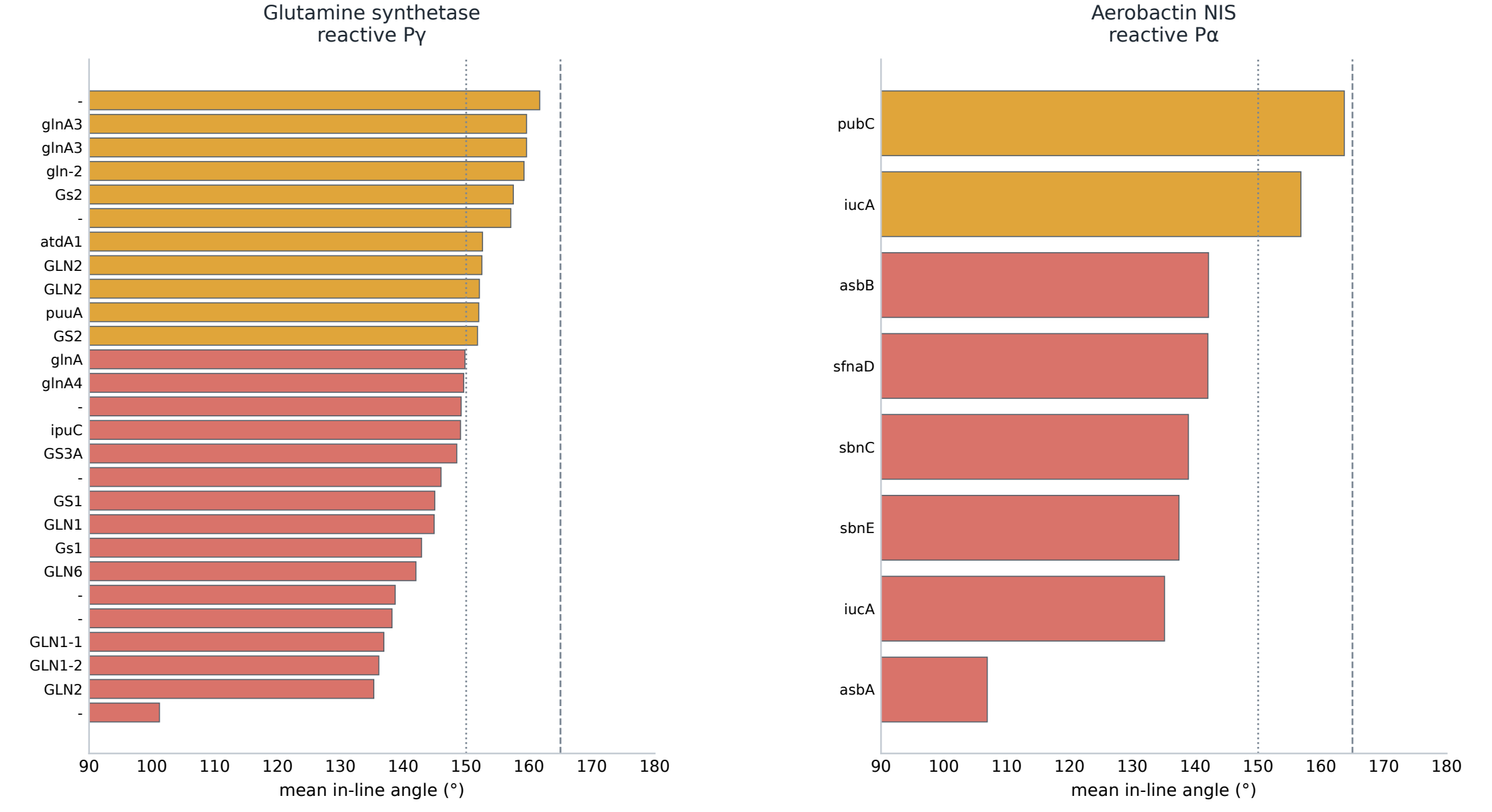
Mechanism-aware step-1 geometry — in-line attack angle, scored vs the correct ATP phosphorus

50 unrestrained Boltz-2 co-folds per enzyme; angle compared only within a mechanistic family. ≥165° strong · 150–165° compatible · <150° weak.